

Python Programming for Digital Forensics and Security Analysis

One of the many uses of the versatile Python programming language is in digital forensics and security analysis. This article covers various aspects like socket programming, port scanning, geo-location and extraction of data from websites like Twitter.



Python is one of the powerful programming languages used in key domains like cloud computing, Big Data analytics, network forensics, mobile app development, Web development and many others. Python has been in use for more than two decades.

Python code follows and provides support for multiple programming paradigms including imperative, functional, procedural and object oriented. Nowadays, Python is widely used for a variety of high performance computing applications by a number of corporate giants including Microsoft, Google, Red Hat, IBM, Amazon and many others. Python is free and open source, and delivers the implementations and interfaces for many other languages and platforms.

Table 1 displays a list of Python implementations, including the support for different platforms and programming models.

Installing Python

Python is available in two versions — Python 3.5 and Python

2.7 (<https://www.python.org>). Either of these can be downloaded, depending upon your requirements and type of application.

Python programming works with the IDE platform on which coding can be done. The command shell interface or IDE supports a Python program that is saved with a .py extension and executed at the command shell interface with the following commands:

```
$ python <filename.py> (For Linux)
```

```
DriveLetter:(\Path-To-Python)>python <filename.py> (For Windows)
```

Table 1

Python implementation	Supporting platform and language
IronPython	.NET Framework
CPython	C
Jython	Java
MicroPython	Microcontrollers
PyPy	Just-In-Time Compiler